

4339 Asset Pricing and Investments - Assignment 2

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1 The CAPM

Consider an economy with a portfolio of risky assets and a risk-free asset in which the CAPM holds. The following table gives the expected return $\mathbb{E}[r]$ and the standard deviation σ of these two assets.

Portfolio	Expected Return	Standard Deviation
Risk free (r_f)	2%	0%
Market (r_M)	8%	20%

1.1 (a) Computation of Beta β for an asset that correlates with the market.

Assume asset A has standard deviation $\sigma_A = 10\%$ and correlation $\rho_{A,M} = 0.5$ with the market. Beta measures the sensitivity of asset A 's returns relative to the market M . It is defined as

$$\beta_A = \frac{\sigma_{A,M}}{\sigma_M^2} \quad \text{where} \quad \sigma_{A,M} = \rho_{A,M} \sigma_A \sigma_M$$

Substituting the values yields

$$\beta_A = \frac{0.5 \cdot 0.1 \cdot 0.2}{0.2^2} = 0.25.$$

Asset A 's returns move one quarter as much as the market's.

1.2 (b) Deriving the risk-aversion coefficient under the CAPM

Under the CAPM all investors hold a combination of the risk-free asset and the tangency portfolio T , which is equal to the market portfolio. Therefore, $\mathbb{E}[r_T] = \mathbb{E}[r_M] = 8\%$ and $\sigma_T = \sigma_M = 20\%$.

The investor puts weight w_M into the market portfolio and weight $1 - w_M$ into the risk-free asset, with the goal to maximize his utility function

$$U = r_f + w_M(\mathbb{E}[r_M] - r_f) - \frac{a}{2}(w_M)^2 \sigma_M^2. \quad (1)$$

Since the risk-free asset has zero variance, the standard deviation of the investor's portfolio is

$$\begin{aligned}\sigma_P &= w_M \sigma_M \\ w_M &= \frac{\sigma_P}{\sigma_M}.\end{aligned}$$

Given that the portfolio standard deviation is $\sigma_P = 0.1$, this results in

$$w_M = \frac{0.10}{0.20} = 0.5 \quad \text{and} \quad w_f = 1 - w_M = 0.5.$$

To find the maximum of the utility formula we differentiate (1) with respect to w_M and set it equal to zero:

$$\begin{aligned}\frac{dU}{dw_M} &= (\mathbb{E}[r_M] - r_f) - a w_M \sigma_M^2 = 0 \\ a &= \frac{(\mathbb{E}[r_M] - r_f)}{w_M \sigma_M^2}\end{aligned}$$

Substituting the values yields

$$a = \frac{0.08 - 0.02}{0.5 \cdot 0.2^2} = \frac{0.06}{0.02} = 3.$$

The investor's risk-aversion coefficient is 3.

2 CAPM with public and private equity

There are two broad classes of risky assets: private equity, P , and public equity, U . Their joint covariance matrix has the block structure

$$\Sigma = \begin{pmatrix} \Sigma_P & \Sigma_{PU} \\ \Sigma_{PU}^\top & \Sigma_U \end{pmatrix}.$$

There are two investors: a private-equity (PE) fund and a public-equity mutual fund. Unless a subexercise imposes an additional restriction, we maintain the standard CAPM assumptions:

- Investors are price takers and choose portfolios using one-period mean–variance preferences.
- Investors have homogeneous beliefs about expected returns, variances, and covariances.
- A risk-free asset exists, and investors can borrow or lend at the common rate r_f .
- Assets are divisible, and there are no taxes, transaction costs, or other trading frictions apart from the restrictions stated in each subexercise.
- The supplies of the risky assets are fixed, so equilibrium market clearing requires investors collectively to hold all outstanding public and private equity.

2.1 (a) Frictionless benchmark

Both the PE fund and the public-equity mutual fund can freely invest in public and private equity. Therefore, they face the same investment opportunity set. Under the standard CAPM assumptions, both investors hold combinations of the risk-free asset and the same tangency portfolio of risky assets.

In equilibrium, market clearing implies that this common tangency portfolio must be the value-weighted portfolio of all risky assets, including both private and public equity. Denote it by

$$r_M = w_P^T r_P + w_U^T r_U,$$

where w_P and w_U are the assets' market-value weights and their elements sum jointly to one. The standard CAPM consequently holds for every public or private asset i :

$$E[r_i] - r_f = \beta_{i,M}(E[r_M] - r_f), \quad \beta_{i,M} = \frac{\text{Cov}(r_i, r_M)}{\text{Var}(r_M)}.$$

Thus, a CAPM using only a public-stock index is generally not the correct model in this benchmark. Such an index omits private equity and is the relevant market portfolio only in the special case in which it is itself mean-variance equivalent to the aggregate portfolio of all risky assets.

The result of this discussion is that the standard CAPM holds, and its relevant market portfolio is the value-weighted portfolio of all private and public equity.

2.2 (b) Investor (demand-side) segmentation

The PE fund can invest only in private equity and the risk-free asset, whereas the equity mutual fund can invest only in public equity and the risk-free asset. Hence, the two investors face different risky investment opportunity sets and choose different tangency portfolios.

The PE fund generates all demand for private equity. It combines the risk-free asset with the tangency portfolio formed from private assets. Because no other investor can hold private equity, market clearing requires this tangency portfolio to equal the value-weighted portfolio of all private equity, denoted by r_{M_P} .

Similarly, the mutual fund generates all demand for public equity. Its tangency portfolio is formed only from public assets, and market clearing requires it to equal the value-weighted portfolio of all public equity, denoted by r_{M_U} .

A separate CAPM therefore holds within each asset class. For every private asset $i \in P$,

$$E[r_i] - r_f = \beta_{i,M_P}(E[r_{M_P}] - r_f), \quad \beta_{i,M_P} = \frac{\text{Cov}(r_i, r_{M_P})}{\text{Var}(r_{M_P})}.$$

For every public asset $j \in U$,

$$E[r_j] - r_f = \beta_{j,M_U}(E[r_{M_U}] - r_f), \quad \beta_{j,M_U} = \frac{\text{Cov}(r_j, r_{M_U})}{\text{Var}(r_{M_U})}.$$

Thus, there is generally no single CAPM with one economy-wide market portfolio that prices both classes. The relevant benchmark for private equity is the private-equity market portfolio, while the relevant benchmark for public equity is the public-equity market portfolio. The cross-covariance block Σ_{PU} does not affect either investor's portfolio choice because neither investor can hold assets from both classes. Consequently, the two classes can have different prices of risk, determined separately by demand and supply within each segmented market.

Therefore, two class-specific CAPMs hold: M_P prices private equity and M_U prices public equity, while no single economy-wide CAPM generally holds.

2.3 (c) Uncorrelated asset classes and imperfect investor segmentation

The PE fund can now invest in both private and public equity, whereas the equity mutual fund remains restricted to public equity. Because $\Sigma_{PU} = 0$, the covariance matrix and its inverse are block diagonal:

$$\Sigma = \begin{pmatrix} \Sigma_P & 0 \\ 0 & \Sigma_U \end{pmatrix}, \quad \Sigma^{-1} = \begin{pmatrix} \Sigma_P^{-1} & 0 \\ 0 & \Sigma_U^{-1} \end{pmatrix}.$$

Let $\mu_P = E[r_P] - r_f \mathbf{1}_P$ and $\mu_U = E[r_U] - r_f \mathbf{1}_U$ denote the vectors of expected excess returns. Mean-variance demand is proportional to $\Sigma^{-1}\mu$. If τ_{PE} and τ_{MF} denote the investors' risk tolerances, their risky-asset demands are therefore

$$x_P^{PE} = \tau_{PE} \Sigma_P^{-1} \mu_P, \quad x_U^{PE} = \tau_{PE} \Sigma_U^{-1} \mu_U,$$

$$x_P^{MF} = 0, \quad x_U^{MF} = \tau_{MF} \Sigma_U^{-1} \mu_U.$$

The zero cross-covariance makes the PE fund's private- and public-equity demands separable: adding public equity does not hedge or alter the demand for private equity, and vice versa.

Let s_P and s_U denote the outstanding supplies of the individual assets in the two classes. Market clearing gives

$$s_P = x_P^{PE} = \tau_{PE} \Sigma_P^{-1} \mu_P,$$

because only the PE fund holds private equity, and

$$s_U = x_U^{PE} + x_U^{MF} = (\tau_{PE} + \tau_{MF}) \Sigma_U^{-1} \mu_U,$$

because both investors hold public equity. Consequently,

$$\mu_P = \frac{1}{\tau_{PE}} \Sigma_P s_P, \quad \mu_U = \frac{1}{\tau_{PE} + \tau_{MF}} \Sigma_U s_U.$$

Within private equity, expected returns thus satisfy a CAPM relative to the value-weighted private-equity market portfolio M_P :

$$E[r_i] - r_f = \beta_{i,M_P}(E[r_{M_P}] - r_f), \quad i \in P.$$

Within public equity, expected returns satisfy a CAPM relative to the value-weighted public-equity market portfolio M_U :

$$E[r_j] - r_f = \beta_{j,M_U}(E[r_{M_U}] - r_f), \quad j \in U.$$

Both investors choose the same relative composition within their public-equity holdings, so their combined holdings make up the public-equity market portfolio. The PE fund alone holds the entire private-equity supply, so its private-equity holdings reproduces the private-equity market portfolio.

In general, there is still no single CAPM based on the value-weighted portfolio of all public and private equity. Private-equity risk is borne only by the PE fund, whereas public-equity risk is shared by both investors. This difference generally produces different equilibrium compensation for risk across the two classes. A economy-wide CAPM would require the two prices of risk to be equal, for which no mechanism exists here.

Following this line of argumentation, separate class-specific CAPMs hold, with M_P as the benchmark for private assets and M_U as the benchmark for public assets, but the aggregate market portfolio does not generally support a standard CAPM.

2.4 (d) Imperfect investor segmentation with correlated asset classes

As in part (c), the PE fund can invest in both private and public equity, while the equity mutual fund can invest only in public equity. The difference is that $\Sigma_{PU} \neq 0$, so public equity can now hedge or amplify the risk of the PE fund's private-equity holdings.

Let x_P^{PE} and x_U^{PE} denote the PE fund's holdings and let x_U^{MF} denote the mutual fund's public-equity holdings. Their mean-variance first-order conditions can be written as

$$\mu_P = a_{PE} (\Sigma_P x_P^{PE} + \Sigma_{PU} x_U^{PE}),$$

$$\mu_U = a_{PE} (\Sigma_{PU}^T x_P^{PE} + \Sigma_U x_U^{PE}) = a_{MF} \Sigma_U x_U^{MF},$$

where μ_P and μ_U are the vectors of expected excess returns and a_{PE} and a_{MF} are the investors' risk-aversion coefficients. Solving the public-equity conditions for the two demands gives

$$x_U^{PE} = \frac{1}{a_{PE}} \Sigma_U^{-1} \mu_U - \Sigma_U^{-1} \Sigma_{PU}^T x_P^{PE},$$

$$x_U^{MF} = \frac{1}{a_{MF}} \Sigma_U^{-1} \mu_U.$$

The mutual fund chooses public assets only for their public-equity risk-return trade-off. The PE fund also uses public assets to hedge its private-equity position, as shown by the additional term involving $\Sigma_{PU}^\top x_P^{PE}$. Therefore, the two investors generally do not hold the same relative weights within public equity.

Market clearing requires

$$s_P = x_P^{PE}, \quad s_U = x_U^{PE} + x_U^{MF}.$$

The public-equity market portfolio is consequently a mixture of two differently composed public portfolios. It need not coincide with the public-only tangency portfolio of the mutual fund or with the PE fund's public equity holdings. Likewise, although the PE fund holds the entire private-equity supply, its optimal private holdings depend on their covariance with its public holdings. The private-equity market portfolio alone is therefore not generally a tangency portfolio either.

The usual CAPM equilibrium argument now breaks down. In the standard CAPM, every investor holds the same risky tangency portfolio, so adding their demands makes that tangency portfolio equal the value-weighted market portfolio. Here, the investors hold different risky portfolios. Their aggregate holdings reproduce asset supply, but aggregation does not produce a common tangency portfolio. Thus, neither the value-weighted public-equity portfolio nor the value-weighted portfolio of all risky assets is generally mean-variance efficient, and neither generally serves as a valid CAPM benchmark.

A portfolio-specific CAPM relation still holds for each investor's own opportunity set. If T_{PE} is the PE fund's tangency portfolio formed from both asset classes, then

$$E[r_i] - r_f = \beta_{i,T_{PE}}(E[r_{T_{PE}}] - r_f), \quad i \in P \cup U.$$

If T_{MF} is the mutual fund's public-only tangency portfolio, then

$$E[r_j] - r_f = \beta_{j,T_{MF}}(E[r_{T_{MF}}] - r_f), \quad j \in U.$$

These are investor-specific tangency-portfolio relations rather than a standard equilibrium CAPM with an observable market portfolio.

Therefore, the standard market CAPM generally does not hold: T_{PE} prices the assets available to the PE fund, T_{MF} prices the public assets available to the mutual fund, and there is no unique market portfolio that prices every asset.

2.5 (e) Listed PE fund

The PE fund is now publicly listed, and its shares trade at net asset value. A share in the fund is therefore a claim on the fund's underlying private- and public-equity holdings. If r_L denotes the return on the listed fund, then

$$r_L = w_{P,L}^\top r_P + w_{U,L}^\top r_U,$$

where $w_{P,L}$ and $w_{U,L}$ are the weights of private and public assets held by the fund.

Because the PE fund can invest in both asset classes, its optimal risky portfolio is the tangency portfolio for the full universe of private and public equity. Listing the fund makes the return on this full-universe tangency portfolio directly available to the mutual fund. Since public stocks were already part of the universe from which that portfolio was formed, we cannot improve the mutual fund's Sharpe ratio by holding them separately.

The mutual fund therefore also chooses the listed PE fund as its risky tangency portfolio and combines it with the risk-free asset. Both investors now hold the same composition of risky wealth. The common tangency-portfolio argument is restored, and market clearing identifies this tangency portfolio with the value-weighted portfolio of all risky assets.

The standard CAPM therefore holds:

$$E[r_i] - r_f = \beta_{i,M}(E[r_M] - r_f), \quad \beta_{i,M} = \frac{\text{Cov}(r_i, r_M)}{\text{Var}(r_M)}.$$

The relevant market portfolio can be described in two equivalent ways:

1. On a look-through basis, it is the value-weighted portfolio of all underlying private and public equity, with every underlying asset counted once.
2. In terms of securities available to the mutual fund, it is the value-weighted portfolio of the listed PE fund and the public equity held outside that fund.

The second description must not add the PE fund's full market capitalization to the full market capitalization of all public companies without an adjustment. Doing so would count the public shares owned inside the PE fund twice. At net asset value, the correctly adjusted securities-level portfolio and the look-through portfolio have exactly the same payoff.

Hence, unlike in part (d), a public-stock benchmark can represent the complete market portfolio if it includes the listed PE fund and is adjusted for its underlying cross-holdings. The assumption that the fund trades at net asset value is important: a discount, premium, management-cost wedge, or other fund-specific friction would break the exact equivalence between the listed claim and its underlying portfolio.

The standard CAPM holds, and the relevant market portfolio is all private and public equity counted once, equivalently represented by the listed PE fund together with public equity held outside the fund.

2.6 (f) Implications for applying the CAPM

Segmentation changes which risks investors can trade and which investors are marginal in each market. Consequently, the appropriate CAPM benchmark need not be the value-weighted portfolio of every risky asset in the economy. It should instead be a mean-variance-efficient portfolio for the opportunity set relevant to the investors who price the assets under consideration.

It is therefore not always a bad idea to use a public-stock index when pricing public equity. Its validity depends on the economic setting:

- In the frictionless benchmark of part (a), investors can hold both asset classes. The relevant market portfolio contains all public and private equity, so a public-only index generally omits priced risk and is not the correct CAPM benchmark.
- Under complete segmentation in part (b), public equity is priced within its own market. The value-weighted public-equity portfolio is the relevant benchmark for public stocks even though it is not the economy-wide wealth portfolio.
- In part (c), the asset classes are uncorrelated. Both investors choose the same relative public-equity weights, so their aggregate public holdings equal the public tangency portfolio. A public-stock index is again an appropriate benchmark for public equity.
- In part (d), private and public returns are correlated. The PE fund uses public assets partly to hedge private-equity risk, while the mutual fund does not. Their public holdings therefore differ, and the aggregate public-stock index need not be a tangency portfolio. In this case, using it as a CAPM benchmark can be misleading.
- In part (e), listing the PE fund at net asset value makes private-equity exposure indirectly tradable. The standard CAPM is restored, but the public benchmark must include the listed PE fund and correctly account for its underlying holdings.

The main point is that which investors price an asset what else they hold, determines whether a proxy is fitting, but not whether the proxy happens to be observable. A public-stock index can price public equity well when it represents the relevant investors' tangency portfolio or spans the omitted risks. It can fail when excluded assets affect those investors' hedging demands or when the index leaves out a material component of their investable aggregate wealth.

We can summarize this by noting that using a public-stock index is not always a bad idea: it is correct when that index is the relevant public-market tangency portfolio or spans the complete market, and it is incorrect when segmentation and cross-asset hedging prevent it from being mean-variance efficient.

3 Case Study

The following scripts in *Section 3* were generated by Claude Opus 5.

First, ensure the required packages are available and set universal formats for the following data analysis and graphics.

```
[12]: import sys, subprocess, importlib

def install_and_import(package, alias=None):
    try:
        module = importlib.import_module(package)
    except ImportError:
        print(f"{package} not found. Installing...")
        subprocess.check_call([sys.executable, "-m", "pip", "install", package])
        module = importlib.import_module(package)
    globals()[alias or package] = module

install_and_import("statsmodels")
```

```

import numpy as np
import pandas as pd
import statsmodels.api as sm
import matplotlib.pyplot as plt

pd.set_option("display.float_format", lambda x: "%.4f" % x)
pd.set_option("display.width", 100)

# Colours (fixed per asset throughout)
COL = {"STOXX": "#2a78d6", "TKA": "#eb6834", "OLE": "#1baf7a"}
NAMES = {"STOXX": "STOXX Europe TMI", "TKA": "ThyssenKrupp", "OLE": "Deoleo"}
plt.rcParams.update({"axes.spines.top": False, "axes.spines.right": False,
                    "axes.grid": True, "grid.alpha": 0.3, "figure.dpi": 110})

```

Next, we load the data from the two provided files and compute the (excess) returns of the three assets.

```

[13]: prices = pd.read_csv("Stockprices.txt", sep="\t")
prices.columns = ["date"] + [c.strip() for c in prices.columns[1:]]
prices["date"] = pd.to_datetime(prices["date"].astype(int).astype(str),
    ↪format="%Y%m") + pd.offsets.MonthEnd(0)
prices = prices.set_index("date")

rf = pd.read_csv("Riskfree.txt", sep="\t").dropna()
rf["date"] = pd.to_datetime(rf["date"].astype(int).astype(str), format="%Y%m")
    ↪+ pd.offsets.MonthEnd(0)
rf = rf.set_index("date")["Return"] / 100          # file is in percent ->
    ↪decimal

rets = prices.pct_change().dropna()                # simple monthly returns
assert rets.index.equals(rf.index), "Return and T-bill dates do not line up"

excess = rets.sub(rf, axis=0)                        # excess returns over the
    ↪T-bill
assets = ["STOXX", "TKA", "OLE"]

print(f"Sample: {excess.index[0]:%b %Y} to {excess.index[-1]:%b %Y}, T =
    ↪{len(excess)} months")
excess.head()

```

Sample: Jan 2006 to Mar 2014, T = 99 months

```

[13]:
      date  STOXX  TKA  OLE
2006-01-31  0.0366  0.3062  0.0289
2006-02-28  0.0188  0.0093 -0.0188
2006-03-31  0.0188  0.1200  0.0303

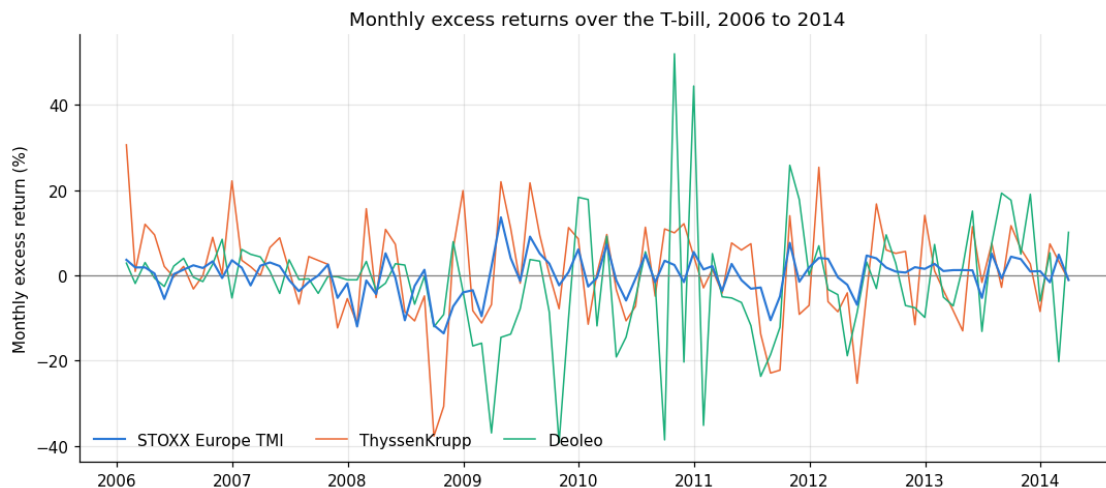
```

```
2006-04-30  0.0046  0.0953 -0.0064
2006-05-31 -0.0552  0.0211 -0.0259
```

The script below outputs an initial graphic to visualize the monthly excess returns over the risk-free (T-bill) rate.

It should be noted that Deoleo's two large positive months in late 2010 (+52% in Oct, +44% in Dec) coincide with its restructuring. We keep this in mind as potentially relevant outliers.

```
[14]: fig, ax = plt.subplots(figsize=(10, 4.5))
      for a in assets:
          ax.plot(excess.index, excess[a] * 100, color=COL[a], lw=1.4 if a == "STOXX"
                  ↪ else 1.0,
                  label=NAMES[a], zorder=3 if a == "STOXX" else 2)
      ax.axhline(0, color="grey", lw=0.8)
      ax.set_ylabel("Monthly excess return (%)")
      ax.set_title("Monthly excess returns over the T-bill, 2006 to 2014")
      ax.legend(loc="lower left", frameon=False, ncol=3)
      plt.tight_layout(); plt.show()
```



3.1 Changing portfolio weights

Background and derivations

SF runs a tactical asset allocation (market timing) strategy around a 60/40 stock/T-bill benchmark, within a 40% to 80% equity range.

Capital allocation: The assets available are a safe asset with return r_f (T-bills) and a risky asset with expected return $\mathbb{E}[r]$ and volatility σ (the STOXX index). A portfolio P is given by weights

w and $1 - w$ on the risky and safe asset:

$$r_P = wr + (1 - w)r_f$$

$$\mathbb{E}[r_P] = r_f + w(\mathbb{E}[r] - r_f) = r_f + w\mathbb{E}[r^e] \quad (r^e \text{ excess return})$$

$$\mathbb{V}ar[r_P] = \mathbb{V}ar[wr + (1 - w)r_f] = \mathbb{V}ar[wr] = w^2 \mathbb{V}ar[r], \quad \sigma_P = w\sigma$$

Optimal capital allocation: A mean-variance investor with risk aversion a maximizes

$$U(w) = \mathbb{E}[r_f + wr^e] - \frac{a}{2} \mathbb{V}ar[r_f + wr^e] = r_f + w\mathbb{E}[r^e] - \frac{a}{2} w^2 \sigma^2.$$

The first-order condition (FOC) gives

$$\underbrace{\mathbb{E}[r^e]}_{\text{marginal benefit}} = \underbrace{aw\sigma^2}_{\text{marginal cost}} \Rightarrow w^* = \frac{1}{a} \frac{\mathbb{E}[r^e]}{\sigma^2}.$$

The optimal weight w^* increases with the risky asset excess return $\mathbb{E}[r^e]$ and decreases with volatility σ and risk aversion a .

Capital Allocation Line: A higher weight w on the risky asset increases excess return and risk:

- Risk premium: $\mathbb{E}[r_P^e] = w\mathbb{E}[r^e]$ (1) - Risk: $\sigma_P = w\sigma \Rightarrow w = \frac{\sigma_P}{\sigma}$ (2)

Equations (1) and (2) imply a relationship between $\mathbb{E}[r_P]$ and σ_P :

$$\mathbb{E}[r_P] = r_f + w\mathbb{E}[r^e] = r_f + \frac{\sigma_P}{\sigma} \mathbb{E}[r^e] = r_f + \sigma_P \frac{\mathbb{E}[r] - r_f}{\sigma} \quad (\text{CAL})$$

with intercept r_f and slope $\frac{\mathbb{E}[r] - r_f}{\sigma}$, the Sharpe ratio.

Sharpe ratio: All portfolios on the CAL have the same Sharpe ratio:

$$S_P = \frac{\mathbb{E}[r_P^e]}{\sigma_P} = \frac{\mathbb{E}[wr^e]}{w\sigma} = \frac{\mathbb{E}[r^e]}{\sigma} = S$$

with S the Sharpe ratio of the risky asset used for capital allocation.

3.1.1 (a) Motivation(s) of Schumpeter's manager for changing portfolio weights

The main motivation for changing the weights is market timing. Schumpeter's manager moves money between the stock index and T-bills depending on where she thinks the market is heading. She overweights equities when she expects the market to rise, as in 2006, in 2012 to 2013, and today at 65%, and moves towards T-bills when she expects a fall, as when she cut the equity share towards 40% in 2007 and 2008. In terms of the optimal weight $w^* = \mathbb{E}[r^e]/(a\sigma^2)$, she acts on the view that the equity premium and market volatility change over time, so a fixed 60% cannot be optimal all the time. When her views are right, the fund earns the equity premium in good years and avoids part of the losses in bad ones. 2008 is the clearest example: the fund lost 26.8%, against 33.3% for the benchmark and 46% for the equity market.

The second motivation is differentiation. Any investment firm can offer a fixed 60/40 portfolio, so SF needs to offer something more to attract investors. The strategy does not have to beat the whole

market to be seen as successful. Beating a conservative 60/40 index/T-bill portfolio is enough, and SF has done that by 0.98% per year since 2006.

The strategy also suits a firm of SF's size. Stock picking needs research capacity that a firm with one investment professional does not have, whereas market timing only requires trading a liquid index and government bills. Trading costs are low (around 7 bp per year on average) and the strategy can be run with a small amount of AUM.

3.1.2 (b) Effect of varying the weights on the reward to risk ratio

Under mean-variance theory the other manager is right. With a risk-free asset and one risky portfolio, every mix of the stock index and T-bills lies on the Capital Allocation Line. A weight w in the index gives an expected excess return of $w \mathbb{E}[r^e]$ and a standard deviation of $w\sigma$, so the Sharpe ratio of the mix is

$$S_P = \frac{w \mathbb{E}[r^e]}{w \sigma} = \frac{\mathbb{E}[r^e]}{\sigma} = S.$$

The weight cancels out. Changing it moves the portfolio up or down the CAL, so the investor takes more or less risk, but the reward per unit of risk stays the same and the line does not get any steeper.

This relies on there being no alpha, with expected returns constant and already reflected in prices. If some information is not priced in, a manager who raises the equity weight ahead of periods with higher expected returns, and lowers it ahead of weaker periods, is effectively switching between CALs with different slopes. The timing strategy can then earn a higher realised Sharpe ratio than any fixed mix such as the 60/40 benchmark. SF's outperformance of 0.98% per year since 2006 fits this story, although it only reflects skill if the timing comes from real information and not from luck.

3.2 The cost in risk of moving 1% from T-bills into stocks

Throughout, returns are excess returns over the T-bill. The T-bill has zero excess return and zero variance, so a portfolio with weights w_i in the risky assets has excess return $r_P^e = \sum_i w_i r_i^e$ and the T-bill weight drops out of all risk calculations.

3.2.1 (a) Annualized standard deviation of excess returns

If monthly excess returns are independent over time with the same variance σ_m^2 , the annual excess return is approximately the sum of the 12 monthly ones. All covariance terms are zero, so

$$\mathbb{V}ar\left[\sum_{t=1}^{12} r_t^e\right] = \sum_{t=1}^{12} \mathbb{V}ar[r_t^e] = 12 \sigma_m^2 \quad \Rightarrow \quad \sigma_{\text{annual}} = \sqrt{12} \sigma_m.$$

```
[15]: sd_m = excess[assets].std()                    # sample SD (ddof = 1)
      sd_a = sd_m * np.sqrt(12)

      q2a = pd.DataFrame({"Monthly SD": sd_m, "Annualized SD": sd_a}).
      ↪rename(index=NAMES)
      q2a.style.format("{:.2%}")
```

```
[15]: <pandas.io.formats.style.Styler at 0x229c7d0c3b0>
```

As expected, the index has the lowest annualized SD (15.73%). The two single stocks are far more volatile, with ThyssenKrupp at 39.70% and Deoleo at 48.41%, roughly 2.5 and 3 times the index.

3.2.2 (b) and 3.2.3 (c) Portfolio standard deviations

Portfolio variance: For $r_P = w_i r_i + w_j r_j$,

$$\text{Var}[r_P] = w_i^2 \sigma_i^2 + 2 w_i w_j \rho_{ij} \sigma_i \sigma_j + w_j^2 \sigma_j^2.$$

- Base and (i) hold only the index, so $\sigma_P = w_M \sigma_M$, i.e. $0.65 \sigma_M$ and $0.66 \sigma_M$. - (ii) and (iii) hold the index plus 1% in stock i :

$$\sigma_P^2 = 0.65^2 \sigma_M^2 + 2 (0.65)(0.01) \rho_{iM} \sigma_i \sigma_M + 0.01^2 \sigma_i^2.$$

With more assets this generalizes to $\sigma_P^2 = \sum_i \sum_j w_i w_j \sigma_{ij} = w^\top \Sigma w$. The code computes the SD directly from the monthly portfolio excess return series, which gives the same result.

```
[16]: portfolios = {
      "Base: 65% STOXX / 35% T-bills": {"STOXX": 0.65, "TKA": 0.00,
      ↪"OLE": 0.00},
      "(i) 66% STOXX / 34% T-bills": {"STOXX": 0.66, "TKA": 0.00,
      ↪"OLE": 0.00},
      "(ii) 65% STOXX + 1% Deoleo / 34% T-bills": {"STOXX": 0.65, "TKA": 0.
      ↪00, "OLE": 0.01},
      "(iii) 65% STOXX + 1% ThyssenKrupp / 34% T-bills": {"STOXX": 0.65, "TKA": 0.
      ↪01, "OLE": 0.00},
      }
      W = pd.DataFrame(portfolios).T[assets]
      W.insert(0, "T-bills", 1 - W.sum(axis=1))

      port_ex = excess[assets] @ W[assets].T          # T x 4 matrix of portfolio
      ↪excess returns
      port_sd = port_ex.std() * np.sqrt(12)

      base = port_sd.iloc[0]
      q2 = W.copy()
      q2["Annualized SD"] = port_sd
      q2["Change vs base (pp)"] = (port_sd - base) * 100
```

```

q2["Change vs base (bp)"] = (port_sd - base) * 1e4

Sigma_a = excess[assets].cov() * 12          # annualized covariance matrix
↳ (used in 2d)

q2.style.format({"T-bills": "{:.0%}", "STOXX": "{:.0%}", "TKA": "{:.0%}", "OLE":
↳ "{:.0%}",
                "Annualized SD": "{:.3%}", "Change vs base (pp)": "{:+.3f}",
                "Change vs base (bp)": "{:+.1f}"})

```

[16]: <pandas.io.formats.style.Styler at 0x229c7d7c9b0>

3.2.3 (d) Risk decomposition

For a small additional position εX added to an existing portfolio R ,

$$\text{Var}(R + \varepsilon X) = \text{Var}(R) + 2\varepsilon \text{Cov}(R, X) + \varepsilon^2 \text{Var}(X).$$

Here R is the current 65% index position, X is the asset added and $\varepsilon = 0.01$. Since $\varepsilon^2 = 0.0001$, the last term is negligible, so the change in risk is driven by the covariance term and not by the asset's own variance. With $R = 0.65 r_M$,

$$\text{Cov}(R, X) = 0.65 \text{Cov}(r_M, r_X) = 0.65 \rho_{XM} \sigma_X \sigma_M,$$

so what matters is $\rho_{XM} \sigma_X$, the part of the asset's volatility that moves with the index. Converting the change in variance into a change in standard deviation ($\Delta \sigma_P \approx \Delta \text{Var} / 2\sigma_P$, with $\sigma_P = 0.65 \sigma_M$) gives

$$\Delta \sigma_P \approx 0.01 \rho_{XM} \sigma_X.$$

The table compares this prediction with the actual change from 2c. The small remaining gap is a second-order effect of the ε^2 term, which only matters for larger positions.

```

[17]: w0 = W.loc[W.index[0], assets]          # base portfolio (65% STOXX)
sig_p = np.sqrt(w0 @ Sigma_a @ w0)
cov_ip = Sigma_a @ w0                        # Cov(r_i, r_P), annualized
corr_iM = excess[assets].corr()["STOXX"]
d = 0.01                                     # size of the trade

decomp = pd.DataFrame({
    "Stand-alone SD": sd_a,
    "Corr with index": corr_iM,
    "Correlated part (rho x SD)": corr_iM * sd_a,
    "Uncorrelated part (SD x sqrt(1-rho^2))": sd_a * np.sqrt(1 - corr_iM**2),
    "Predicted change, 0.01 x rho x SD (bp)": d * cov_ip / sig_p * 1e4,
})
# map each asset to its 2c portfolio: (i) index, (ii) Deoleo, (iii) ThyssenKrupp
chg_2c = pd.Series(q2["Change vs base (bp)"].iloc[1:].values, index=["STOXX",
↳ "OLE", "TKA"])

```

```
decomp["Actual change, +1% (bp)"] = chg_2c
decomp.rename(index=NAMES).T
```

```
[17]:
```

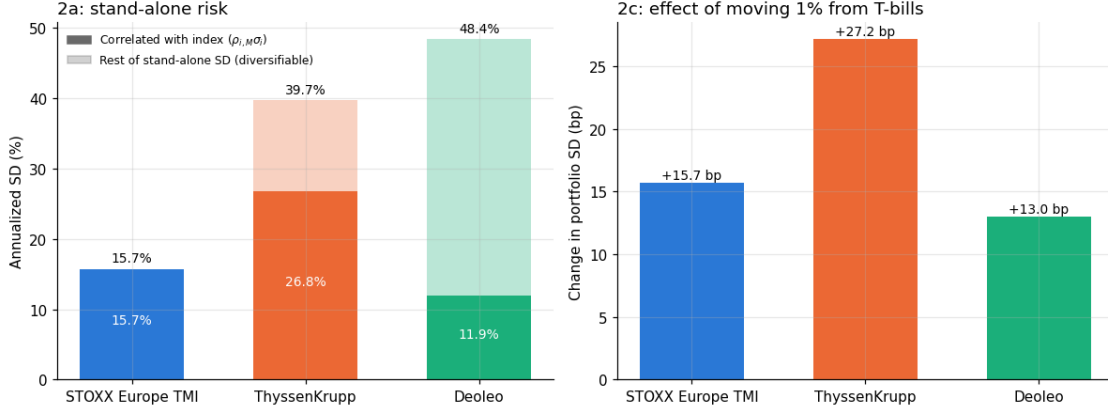
	STOXX Europe TMI	ThyssenKrupp	Deoleo
Stand-alone SD	0.1573	0.3970	0.4841
Corr with index	1.0000	0.6741	0.2461
Correlated part (rho x SD)	0.1573	0.2676	0.1191
Uncorrelated part (SD x sqrt(1-rho^2))	0.0000	0.2932	0.4692
Predicted change, 0.01 x rho x SD (bp)	15.7272	26.7608	11.9140
Actual change, +1% (bp)	15.7272	27.1705	12.9778

The left chart splits each stand-alone SD into the part correlated with the index, $\rho_{iM}\sigma_i$, and the rest. The right chart shows the actual change in portfolio SD from 2c.

```
[18]: fig, axes = plt.subplots(1, 2, figsize=(11, 4.2))
lab = [NAMES[a] for a in assets]
x = np.arange(len(assets))

ax = axes[0]
corr_part = (corr_iM * sd_a) * 100
unc_part = (sd_a - corr_iM * sd_a) * 100
ax.bar(x, corr_part, color=[COL[a] for a in assets], width=0.6,
      label="Correlated with index ( $\rho_{iM}\sigma_i$ )")
ax.bar(x, unc_part, bottom=corr_part, color=[COL[a] for a in assets], alpha=0.3,
      label="Rest of stand-alone SD")
for i, a in enumerate(assets):
    ax.text(i, sd_a[a] * 100 + 1, f"{sd_a[a]:.1%}", ha="center", fontsize=9)
    ax.text(i, corr_part[a] / 2, f"{corr_part[a]:.1f}%", ha="center",
      fontsize=9, color="white")
ax.set_xticks(x, lab); ax.set_ylabel("Annualized SD (%)")
from matplotlib.patches import Patch
ax.set_title("2a: stand-alone risk", loc="left")
ax.legend(handles=[Patch(color="dimgray", label="Correlated with index",
      label="Correlated with index ( $\rho_{iM}\sigma_i$ )",
      Patch(color="dimgray", alpha=0.3, label="Rest of stand-alone",
      label="Rest of stand-alone SD (diversifiable)"),
      frameon=False, fontsize=8, loc="upper left")

ax = axes[1]
chg = chg_2c[assets].values
ax.bar(x, chg, color=[COL[a] for a in assets], width=0.6)
for i, v in enumerate(chg):
    ax.text(i, v + 0.3, f"{v:.1f} bp", ha="center", fontsize=9)
ax.set_xticks(x, lab); ax.set_ylabel("Change in portfolio SD (bp)")
ax.set_title("2c: effect of moving 1% from T-bills", loc="left")
plt.tight_layout(); plt.show()
```



3.2.4 (d) Discussion of 3.2.3 (c) in light of the results from 3.2.1 (a)

The stand-alone standard deviations from 2a suggest that the STOXX index is the least risky asset and Deoleo the most risky. The results in 2c show something different: adding 1% of Deoleo raises the portfolio standard deviation the least, by 13.0 bp, against 15.7 bp for the index and 27.2 bp for ThyssenKrupp. The risk an asset adds to a portfolio depends on how it moves with the rest of the portfolio, which here is 65% the STOXX index, and much less on its own volatility.

Deoleo's correlation with the index is only 0.25, so just 11.9% of its 48.4% volatility moves with the portfolio. The rest is idiosyncratic and gets diversified away when the position is small. ThyssenKrupp has a correlation of 0.67, so 26.8% of its 39.7% volatility moves with the index, which makes it the most expensive addition in terms of risk.

The risk decomposition above shows this directly. Adding a small amount of asset i changes the portfolio standard deviation by $\text{Cov}[r_i, r_P]/\sigma_P = \rho_{iM}\sigma_i$ per unit of weight, so a 1% trade should add roughly $0.01\rho_{iM}\sigma_i$. That gives 15.7, 26.8 and 11.9 bp, close to the actual numbers. The remaining gap (about 1 bp for Deoleo and 0.4 bp for ThyssenKrupp) comes from the $\varepsilon^2 \text{Var}(X)$ term, which grows with the square of the position, so Deoleo's advantage would shrink if the fund took a larger stake.

3.3 Betas and security characteristic lines

3.3.1 (a) Index-model regressions

For each stock, with the STOXX index as the market proxy,

$$r_{i,t}^e = \alpha_i + \beta_i r_{M,t}^e + \epsilon_{i,t}, \quad \mathbb{E}[\epsilon_i] = 0, \quad \text{Cov}[\epsilon_i, r_M^e] = 0.$$

Minimizing the sum of squared residuals gives

$$\hat{\beta}_i = \frac{\widehat{\text{Cov}}[r_i^e, r_M^e]}{\widehat{\text{Var}}[r_M^e]} = \hat{\rho}_{iM} \frac{\hat{\sigma}_i}{\hat{\sigma}_M}, \quad \hat{\alpha}_i = \bar{r}_i^e - \hat{\beta}_i \bar{r}_M^e,$$

so beta is the correlation with the market scaled by relative volatility.

Because ϵ_i is uncorrelated with the market,

$$\text{Var}[r_i^e] = \underbrace{\beta_i^2 \text{Var}[r_M^e]}_{\text{systematic}} + \underbrace{\sigma^2(\epsilon_i)}_{\text{idiosyncratic}}, \quad R^2 = \frac{\beta_i^2 \text{Var}[r_M^e]}{\text{Var}[r_i^e]} = \rho_{iM}^2.$$

The systematic SD $\beta_i \sigma_M$ equals $\rho_{iM} \sigma_i$, the correlated part from 2d, and the residual SD in the table is $\sigma(\epsilon_i)$.

Note on hypothesis tests: Two nulls are of interest. $H_0 : \alpha_i = 0$ is the CAPM prediction of no abnormal average excess return. For beta, $H_0 : \beta_i = 0$ only asks whether the stock has any market exposure. The more meaningful benchmark is $H_0 : \beta_i = 1$, since the market portfolio has a beta of one by definition, so this tests whether the stock is more or less exposed than the market:

$$t_\alpha = \frac{\hat{\alpha}_i}{se(\hat{\alpha}_i)}, \quad t_{\beta=1} = \frac{\hat{\beta}_i - 1}{se(\hat{\beta}_i)}.$$

```
[19]: X = sm.add_constant(excess["STOXX"].rename("Market excess return"))
models = {s: sm.OLS(excess[s], X).fit() for s in ["TKA", "OLE"]}

for s, m in models.items():
    print(f"\n{'=' * 30} {NAMES[s]} {'=' * 30}")
    print(m.summary())
```

```
===== ThyssenKrupp =====
                        OLS Regression Results
=====
```

Dep. Variable:	TKA	R-squared:	0.454		
Model:	OLS	Adj. R-squared:	0.449		
Method:	Least Squares	F-statistic:	80.78		
Date:	Thu, 01 Oct 2026	Prob (F-statistic):	2.05e-14		
Time:	17:06:15	Log-Likelihood:	104.48		
No. Observations:	99	AIC:	-205.0		
Df Residuals:	97	BIC:	-199.8		
Df Model:	1				
Covariance Type:	nonrobust				

```
=====
```

	coef	std err	t	P> t	[0.025
0.975]					

const	0.0083	0.009	0.966	0.336	-0.009
0.025					
Market excess return	1.7016	0.189	8.988	0.000	1.326
2.077					

```
=====
```

Omnibus:	4.873	Durbin-Watson:	1.767
Prob(Omnibus):	0.087	Jarque-Bera (JB):	4.190
Skew:	0.445	Prob(JB):	0.123
Kurtosis:	3.475	Cond. No.	22.1

=====

Notes:

[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.

===== Deoleo =====

OLS Regression Results

=====

Dep. Variable:	OLE	R-squared:	0.061
Model:	OLS	Adj. R-squared:	0.051
Method:	Least Squares	F-statistic:	6.254
Date:	Thu, 01 Oct 2026	Prob (F-statistic):	0.0141
Time:	17:06:15	Log-Likelihood:	57.948
No. Observations:	99	AIC:	-111.9
Df Residuals:	97	BIC:	-106.7
Df Model:	1		
Covariance Type:	nonrobust		

=====

	coef	std err	t	P> t	[0.025
0.975]					

const	-0.0183	0.014	-1.337	0.184	-0.045
0.009					
Market excess return	0.7575	0.303	2.501	0.014	0.156
1.359					

=====

Omnibus:	12.932	Durbin-Watson:	2.245
Prob(Omnibus):	0.002	Jarque-Bera (JB):	40.648
Skew:	0.152	Prob(JB):	1.49e-09
Kurtosis:	6.124	Cond. No.	22.1

=====

Notes:

[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.

```
[20]: from scipy import stats
```

```
rows = {}
sd_M_a = sd_a["STOXX"]
```

```

for s, m in models.items():
    resid_sd_a = np.sqrt(m.mse_resid * 12)
    rows[NAMES[s]] = {
        "alpha (monthly)": m.params["const"],
        "alpha (annualized)": m.params["const"] * 12,
        "t(alpha)": m.tvalues["const"],
        "p-value (alpha = 0)": m.pvalues["const"],
        "beta": m.params["Market excess return"],
        "SE(beta)": m.bse["Market excess return"],
        "t(beta = 0)": m.tvalues["Market excess return"],
        "t(beta = 1)": (m.params["Market excess return"] - 1) / m.bse["Market_
↪excess return"],
        "p-value (beta = 1)": 2 * stats.t.sf(abs((m.params["Market excess_
↪return"] - 1) / m.bse["Market excess return"]), m.df_resid),
        "95% CI beta": "{:.2f}, {:.2f}".format(*m.conf_int().loc["Market_
↪excess return"]),
        "R-squared": m.rsquared,
        "Total SD (ann.)": sd_a[s],
        "Systematic SD = beta x sd_M (ann.)": m.params["Market excess return"]_
↪* sd_M_a,
        "Residual SD (ann.)": resid_sd_a,
    }
q3 = pd.DataFrame(rows)
q3

```

[20]:	ThyssenKrupp	Deoleo
alpha (monthly)	0.0083	-0.0183
alpha (annualized)	0.0991	-0.2195
t(alpha)	0.9659	-1.3366
p-value (alpha = 0)	0.3365	0.1845
beta	1.7016	0.7575
SE(beta)	0.1893	0.3029
t(beta = 0)	8.9880	2.5009
t(beta = 1)	3.7058	-0.8004
p-value (beta = 1)	0.0004	0.4254
95% CI beta	[1.33, 2.08]	[0.16, 1.36]
R-squared	0.4544	0.0606
Total SD (ann.)	0.3970	0.4841
Systematic SD = beta x sd_M (ann.)	0.2676	0.1191
Residual SD (ann.)	0.2947	0.4716

Security characteristic lines

```

[21]: fig, axes = plt.subplots(1, 2, figsize=(12, 5), sharex=True, sharey=True)
xm = excess["STOXX"] * 100
xs = np.linspace(xm.min() - 1, xm.max() + 1, 50)

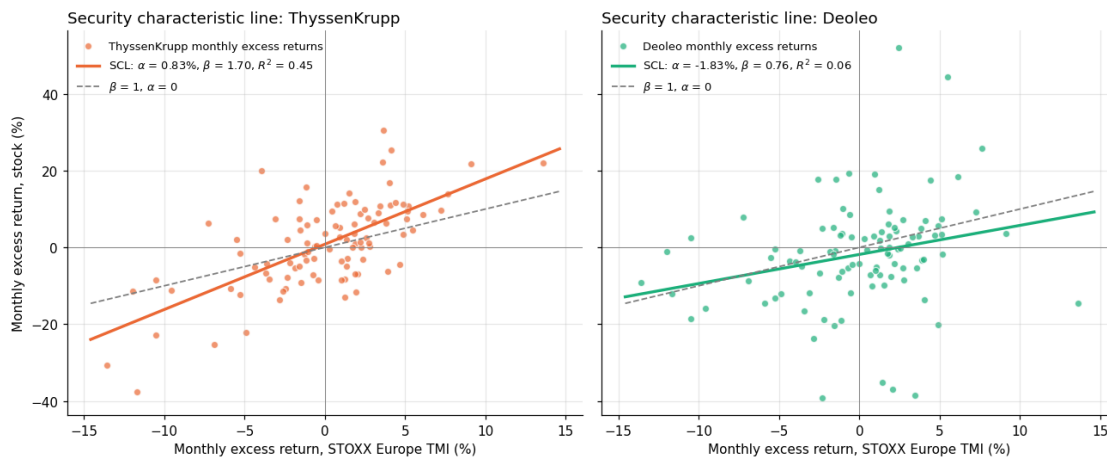
for ax, s in zip(axes, ["TKA", "OLE"]):

```

```

m = models[s]
a, b = m.params["const"] * 100, m.params["Market excess return"]
ax.scatter(xm, excess[s] * 100, s=28, color=COL[s], alpha=0.7,
           edgecolor="white", linewidth=0.8, label=f"{NAMES[s]} monthly_
↳excess returns")
ax.plot(xs, a + b * xs, color=COL[s], lw=2.2,
       label=f"SCL:  $\alpha$  = {a:.2f}%,  $\beta$  = {b:.2f},  $R^2$  = {m.
↳rsquared:.2f}")
ax.plot(xs, xs, color="grey", ls="--", lw=1.2, label=f" $\beta$  = 1,
↳ $\alpha$  = 0")
ax.axhline(0, color="grey", lw=0.6); ax.axvline(0, color="grey", lw=0.6)
ax.set_title(f"Security characteristic line: {NAMES[s]}", loc="left")
ax.set_xlabel("Monthly excess return, STOXX Europe TMI (%)")
ax.legend(loc="upper left", frameon=False, fontsize=8.5)
axes[0].set_ylabel("Monthly excess return, stock (%)")
plt.tight_layout(); plt.show()

```



Robustness: Deoleo's beta without the two restructuring months in late 2010.

```

[22]: mask = ~excess.index.isin(pd.to_datetime(["2010-10-31", "2010-12-31"]))
m_ex = sm.OLS(excess.loc[mask, "OLE"], X.loc[mask]).fit()
print(f"Deoleo beta, full sample: {models['OLE'].params.iloc[1]:.
↳3f} (R2 = {models['OLE'].rsquared:.3f})")
print(f"Deoleo beta, excl. Oct & Dec 2010: {m_ex.params.iloc[1]:.3f} (R2 =
↳{m_ex.rsquared:.3f})")

```

Deoleo beta, full sample: 0.758 (R2 = 0.061)

Deoleo beta, excl. Oct & Dec 2010: 0.577 (R2 = 0.047)

Discussion. Question 3.3.1 (a)

The regressions give a beta of 1.70 for ThyssenKrupp and 0.76 for Deoleo. ThyssenKrupp's beta is precisely estimated, with a 95% confidence interval of 1.33 to 2.08, and the market explains 45% of

its variance. Deoleo's beta is much less precise (0.16 to 1.36) and the market explains only 6% of its variance, which is visible in the SCL plot where most points lie far from the fitted line. Neither alpha is statistically different from zero (t-statistics of 0.97 and -1.34), which is consistent with the CAPM prediction of no abnormal return. Testing against a beta of one gives a different picture for the two stocks. ThyssenKrupp's beta is significantly above one ($t = 3.71$), so it is clearly more exposed to the market than the index. For Deoleo $t = -0.80$, so we cannot reject that its true beta is one, even though the point estimate is well below it.

3.3.2 (b) How these findings support the results from question 2

The betas provide important context and explain the results from question 2. ThyssenKrupp's beta is well above 1, so putting 1% in TKA adds roughly 1.7 times the market risk of putting 1% in the index. Deoleo's beta is below 1, so 1% in Deoleo adds less risk than 1% in the index, even though its stand-alone SD is the highest of the three. This is in line with information provided in the case. ThyssenKrupp is a cyclical steel and industrial company whose earnings depend heavily on European car production and the business cycle. Deoleo sells a basic food product with stable demand, and most of its large price swings came from its own problems, such as the debt taken on to buy Bertolli, the removal of its management and the 2010 restructuring. The low R^2 for Deoleo points the same way, since most of its variance is idiosyncratic and diversifies away inside a large portfolio.

For a small position in a diversified portfolio, stand-alone volatility is therefore the wrong measure of risk and beta is the right one, which is also why only beta risk is priced under the CAPM. There are some caveats. The betas are estimated with error and for Deoleo its beta is not significantly different from one. This makes the central claim that it adds less risk than the index rely on a single and possibly inaccurate point estimate. The betas can also change over time (Deoleo's beta moves noticeably when the 2010 restructuring months are left out), and for larger positions idiosyncratic risk starts to matter again.